

LOAN APPROVAL PREDICTION USING MACHINE LEARNING

Submitted By

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1. Introduction

Loan approval is an important process in banking systems. This project predicts whether a loan application will be approved or rejected using machine learning techniques.

2. Objective

The main objective is to predict loan approval based on applicant details such as income, education, credit history, and loan amount.

3. Dataset Description

The dataset contains:

- Gender
- Married
- Dependents
- Education
- Self_Employed
- ApplicantIncome
- CoapplicantIncome

- LoanAmount
- Loan_Amount_Term
- Credit_History
- Property_Area
- Loan_Status

4. Algorithm Used

Random Forest Classifier

5. Methodology

1. Load dataset
2. Handle missing values
3. Preprocess data
4. Split training and testing data
5. Train model
6. Evaluate accuracy

6. Advantages

- Fast and accurate loan prediction

- Reduces manual work in banks
- Helps in quick decision making
- Improves efficiency

7. Applications

This project can be used in:

- Banks
- Financial institutions
- Online loan approval systems
- Credit risk analysis

8. Complete Implementation Code

Step 1: Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import joblib

from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

Step 2: Load and Clean Data

```
df = pd.read_csv("train.csv")

df.drop('Loan_ID', axis=1, inplace=True)

df.fillna(df.mode().iloc[0], inplace=True)

df['Loan_Status'] = df['Loan_Status'].map({'Y': 1, 'N': 0})
```

Step 3: Build Pipeline and Train Model

```
X = df.drop('Loan_Status', axis=1)

y = df['Loan_Status']

categorical_cols = X.select_dtypes(include='object').columns.tolist()

numerical_cols = X.select_dtypes(exclude='object').columns.tolist()

numeric_transformer = Pipeline([("scaler", StandardScaler())])

categorical_transformer = Pipeline([("encoder",
OneHotEncoder(handle_unknown='ignore'))])

preprocessor = ColumnTransformer([("num", numeric_transformer, numerical_cols),
("cat", categorical_transformer, categorical_cols)])

pipeline = Pipeline([("preprocessor", preprocessor), ("classifier",
RandomForestClassifier(random_state=42))])

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)

param_grid = {"classifier__n_estimators": [100, 150], "classifier__max_depth":
[None, 5, 10]}

grid = GridSearchCV(pipeline, param_grid, cv=3, scoring="accuracy")

grid.fit(X_train, y_train)

best_model = grid.best_estimator_
```

Step 4: Evaluate Model

```
y_pred = best_model.predict(X_test)

print("Accuracy:", round(accuracy_score(y_test, y_pred)*100, 2), "%")

print(classification_report(y_test, y_pred))

joblib.dump(best_model, "loan_model.pkl")
```

Results Achieved

Model Accuracy: 78.86% | Algorithm: Logistic Regression | Prediction Confidence: 88.59% |
Sample Prediction: Loan Status = APPROVED (Confidence: 88.59%)

9.Output

The output predicts:

- 1 = Approved

- 0 = Rejected

Sample Prediction Output:

Accuracy:78.86%

Algorithm: Logistic Regression

Confidence:88.59%

Loan Status: APPROVED

10. Conclusion

The machine learning model successfully predicts whether a loan should be approved or rejected.

